Berke Uğur Aksakal

Cham, Bavaria, Germany (relocating to Regensburg, Oct 2026) | berkeuguraksakal17@gmail.com | +49 1551 08792 91 | berkeuguraksakal.com | github.com/BUAksakal | linkedin.com/in/berke-ugur-aksakal

Available up to 20h/week during the semester (student work permit), full-time during semester breaks

Profile

Master's student in Artificial Intelligence for Smart Sensors and Actuators with a background in Electrical and Electronics Engineering. Hands-on experience across the full ML stack – from real-time object detection and **LangGraph**-orchestrated agents to sensor-to-actuator control loops on physical hardware. Comfortable building data pipelines, iterating on models in simulation, and taking projects from a rough baseline to something that actually works in the real world.

Work Experience

Engineer Intern — Aug 2023 – Jan 2024 (6 months)
NISO Yazılım Teknolojileri A.Ş., İzmir, Türkiye

- Contributed to AI pipelines including YOLO-based detection and **LiDAR-SLAM** workflows for real-time perception; benchmarked YOLOv3 against YOLOv8 for traffic light detection, presented at IEEE TUAC 2023 (selected, undergraduate track).

Engineer Intern — Aug 2024 – Sep 2024 (2 months)
Baylan Ölçü Aletleri Water & Energy Meters, İzmir, Türkiye

- Developed a Python-based desktop tool for real-time device monitoring, data processing, and visualization.

Projects

Journey Resilience Agent — github.com/BUAksakal/journey-resilience-agent
Live transfer-risk monitoring for German rail journeys

- Built a stateful **LangGraph** agent that monitors live German rail journeys via the official **DB Timetables API** and triggers automatic replanning; trained a **LightGBM** quantile regression model on 1.87M delay records (q90 calibrated to 90.7% coverage) to predict missed-transfer probability, served through a **FastAPI** backend with a deployed live web demo.

Low-Altitude Aerial Object Detection (Ariel Project) — github.com/BUAksakal/low-altitude-aerial-detection
Academic-industry collaboration: TH Ingolstadt, Fraunhofer IVI, and TH Deggendorf

- **Led a 4-person team** through the full data pipeline – 4,776 custom annotations (2,227 frames, 4 scenes) in **Roboflow**, fine-tuning **YOLOv8n** in **PyTorch** to **mAP@0.5 = 0.918** (mAP@0.5:0.95 = 0.850), F1 = 0.94, at **2.3ms inference**; exported to **ONNX** for edge/drone deployment.

AI-Based Vision-Controlled Robotic Hand (Bachelor's Final Thesis)
github.com/BUAksakal/ai-vision-robotic-hand

- Built a real-time perception-to-actuator control loop on physical hardware using **Python**, **OpenCV**, **YOLOv8**, and **Arduino (C++)** with **serial communication** – model output driving a real robot, not just a simulation.

Reactive LiDAR Navigation (ROS + Gazebo) — github.com/BUAksakal/lidar-slam-mapping-system

- Built a **ROS** node processing 2D LiDAR sensor data into real-time velocity commands for reactive obstacle avoidance, tested and iterated in **Gazebo** simulation before hardware deployment.

Education

Master's Degree in Artificial Intelligence for Smart Sensors and Actuators — 2025 – Present
Deggendorf Institute of Technology, Germany

Bachelor's Degree in Electrical and Electronics Engineering — 2020 – 2025
Afyon Kocatepe University, Afyon, Türkiye

Skills

Programming: Python, C++, JavaScript, Dart (Flutter), HTML/CSS, MATLAB

AI, ML & Agents: LangGraph, LightGBM, PyTorch, TensorFlow, YOLO (v3/v8), OpenCV, scikit-learn, Roboflow, Fine-Tuning

Robotics & Control: ROS, Gazebo, Kinematics, Motion Planning, EKF / State Estimation, Serial Communication

Tools & Platforms: FastAPI, ONNX, Linux, Git, Arduino, NumPy, Pandas

Languages

Turkish (Native)
German
English (C1)